

Actionable Runbook: Deployment Gotchas

This execution guide covers the final three critical deployment traps: **Secrets Management**, **Persistent Uploads**, and **Port Collisions**. Follow these steps when installing your Personal Loan Tracker on a new laptop, cloud server, or Raspberry Pi to ensure data persists and containers boot flawlessly.

Trap 4: The "Ignored Secrets" Crash (.env file)

Symptom: Your FastAPI backend or Postgres database crashes immediately upon startup on a new system with an error about "missing password" or "invalid credentials."

Step-by-Step Fix

Why this happens: For security, your `.gitignore` prevents the `.env` file (which holds real passwords) from being uploaded to GitHub. When you clone the project to a new PC, the file simply doesn't exist.

1 Duplicate the Example Environment File

You must manually create the `.env` file from the safe template.

```
tony@Jarvis:~/docker/loan-tracker$ cp .env.example .env
```

2 Inject Real Credentials

Open the new `.env` file and replace the placeholder values with your actual secure passwords.

```
tony@Jarvis:~/docker/loan-tracker$ nano .env # Change
POSTGRES_PASSWORD=your_secure_password # Change
JWT_SECRET_KEY=your_long_random_string
```

Trap 5: The "Vanishing Attachments" Bug

Symptom: You upload a PDF to a loan. Later, you restart the backend container or update its code, and suddenly the PDF is permanently gone.

Step-by-Step Fix

1 Mount a Persistent Volume for Uploads

We must explicitly map your host laptop's uploads / folder to the container. Otherwise, the container saves PDFs in its temporary memory, which wipes clean on restart.

 `docker-compose.yml`

```
backend:

  build: ./backend

  volumes:

-   # (Make sure source code bind mounts are disabled in prod)

+   # Map ONLY the uploads folder so PDFs survive reboots!

+   - ./uploads:/app/uploads

  depends_on:

    - db
```

2 Grant Write Permissions (Linux/Ubuntu)

The internal Docker user must be allowed to write files into your laptop's folder.

```
# Grant the container permission to write inside the uploads directory
tony@Jarvis:~/docker/loan-tracker$ chmod -R 777 ./uploads
```

Trap 6: The Port Collision Trap

Symptom: When running `docker-compose up`, it completely fails with a red error: `Bind for 0.0.0.0:5432 failed: port is already allocated.`

Step-by-Step Fix

Why this happens: The new server you are deploying to is *already* running another database on port 5432, or another web server on port 8090. Two apps cannot share the same front door.

1 Shift the External Port in Docker-Compose

You never change the internal code. You only change the **left side** of the port mapping in Docker-Compose to an unused port (like 5433 or 8095).

 `docker-compose.yml`

```
db:
```

```
  image: postgres:15
```

```
  ports:
```

```
-   - "5432:5432" # Host Port : Container Port
```

```
+   - "5433:5432" # Shift host port to 5433 to avoid collision
```

```
nginx:
```

```
  image: nginx:latest
```

```
  ports:
```

```
-   - "8090:80"
```

```
+   - "8095:80"   # Shift host port to 8095 to avoid collision
```

2 Reboot the Stack

```
tony@Jarvis:~/docker/loan-tracker$ docker-compose up -d
```

How to test: You should now be able to access your app in the browser using the new port, e.g., `http://192.168.x.x:8095`, completely avoiding the collision!